Decision Trees



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Decision Trees

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Learning Objectives

In this module, we will learn about decision trees:

- Introduction
- Examples
- Univariate and Multivariate Trees
- Classification Trees

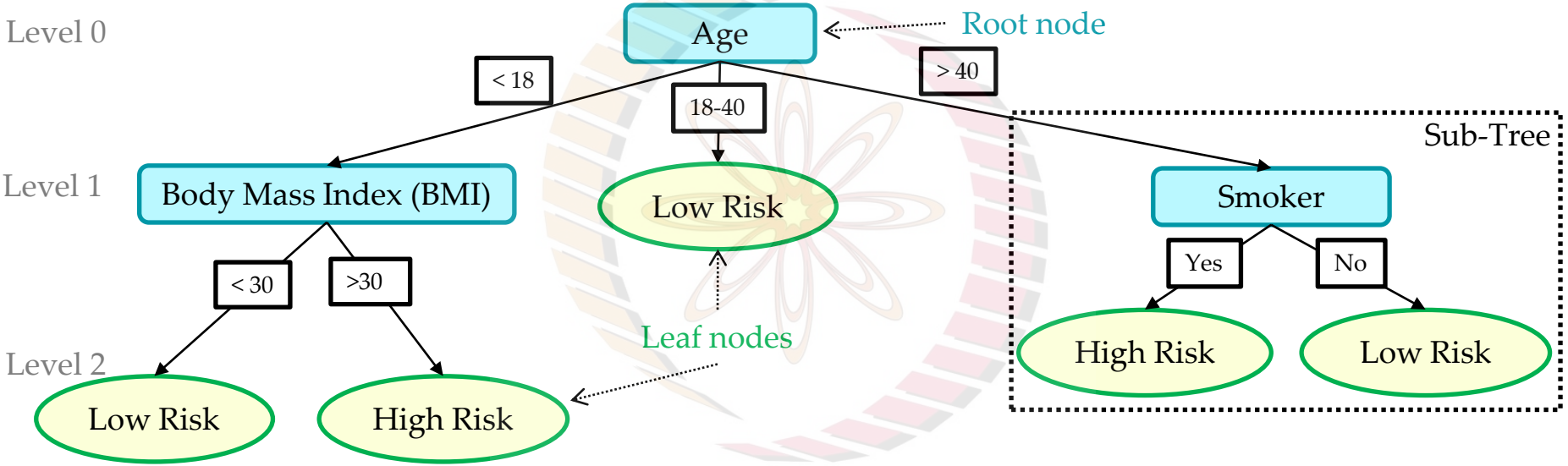
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Introduction

- A type of non-parametric, hierarchical model for supervised learning tasks that makes a sequence of decisions based on the values of the input features
- Composed of internal decision nodes and terminal leaves with final decision (e.g., class label or numeric value)
- Progressively splits the training set into smaller subsets
- Splitting criterion [or attribute selection measure (ASM)]: Gini impurity, entropy impurity, variance impurity, etc.
- Stopping criterion (to stop further splitting): maximum depth, minimum number of samples per leaf, etc.
- Used for classification and regression tasks in ML

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Decision Trees - Example





Task: Classification of a subject into the level of risk of heart attack

High Risk= (BMI>30 AND Age<18) OR (Smoker='Yes' AND Age>40)

Decision Trees - Example

Examples described by attribute values (Boolean, discrete, continuous, etc.)

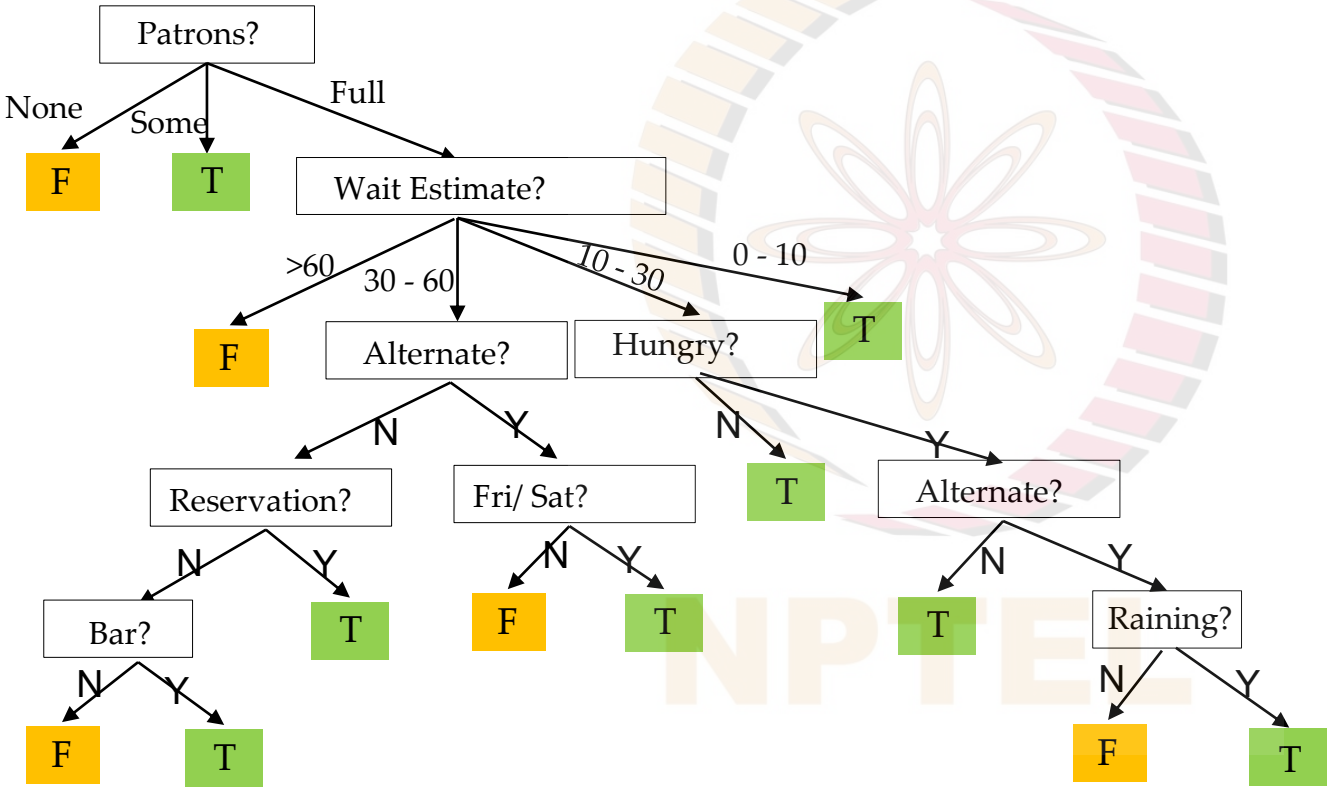
E.g., situations where-- I will wait OR I will not wait:

Example	Attributes										Target
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*Will Wait*
X_1	*T*	*F*	*F*	*T*	*Some*	*$$$*	*F*	*T*	*French*	*0–10*	*T*
X_2	*T*	*F*	*F*	*T*	*Full*	*$*	*F*	*F*	*Thai*	*30–60*	*F*
X_3	*F*	*T*	*F*	*F*	*Some*	*$*	*F*	*F*	*Burger*	*0–10*	*T*
X_4	*T*	*F*	*T*	*T*	*Full*	*$*	*T*	*F*	*Thai*	*10–30*	*T*
X_5	*T*	*F*	*T*	*F*	*Full*	*$$$*	*F*	*T*	*French*	*>60*	*F*
X_6	*F*	*T*	*F*	*T*	*Some*	*$$*	*T*	*T*	*Italian*	*0–10*	*T*
X_7	*F*	*T*	*F*	*F*	*None*	*$*	*T*	*F*	*Burger*	*0–10*	*F*
X_8	*F*	*F*	*F*	*T*	*Some*	*$$*	*T*	*T*	*Thai*	*0–10*	*T*
X_9	*F*	*T*	*T*	*F*	*Full*	*$*	*T*	*F*	*Burger*	*>60*	*F*
X_{10}	*T*	*T*	*T*	*T*	*Full*	*$$$*	*F*	*T*	*Italian*	*10–30*	*F*
X_{11}	*F*	*F*	*F*	*F*	*None*	*$*	*F*	*F*	*Thai*	*0–10*	*F*
X_{12}	*T*	*T*	*T*	*T*	*Full*	*$*	*F*	*F*	*Burger*	*30–60*	*T*

Artificial Intelligence, spring 2013, Peter Ljunglöf; based on AIMA Slides © Stuart Russel and Peter Norvig, 2004

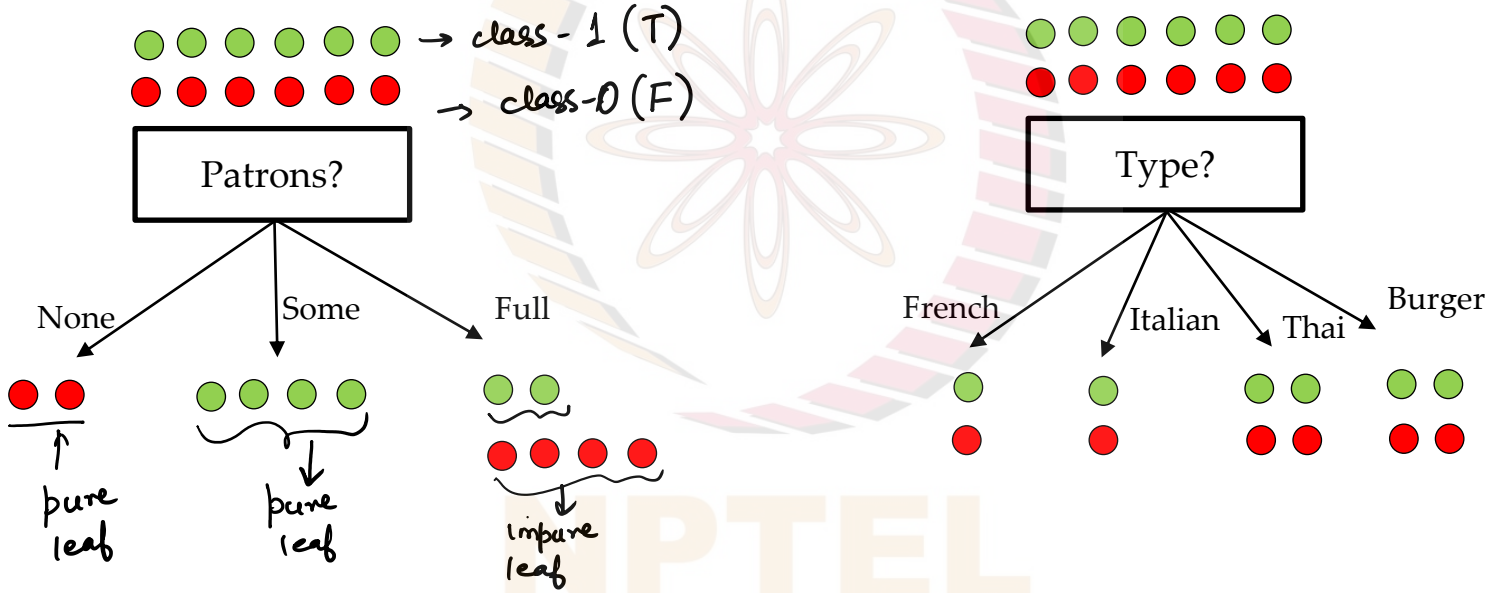
*$Alt(ernate), Fri(day), Hun(gry), Pat(rons), Res(ervation), Est(imated\ waiting\ time)$

Decision Trees - Example



Decision Trees - Example

- A good attribute splits the samples into subsets that are 'all positive' or 'all negative'.



'Patrons' is a better choice—
it gives information about the classification

Types

- Based on the number of features per node, the decision trees are of two types:

Univariate Trees	Multivariate Trees
Only one feature in one node	Multiple features within the same node

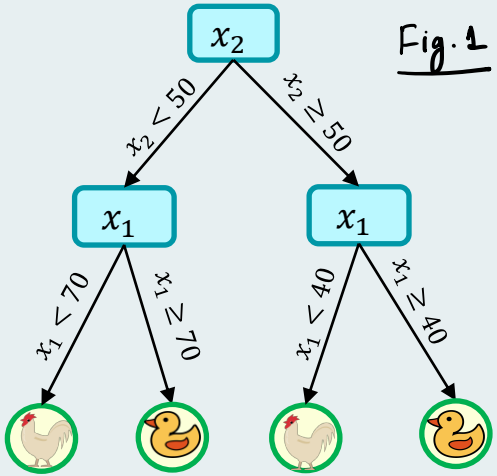


Fig. 1

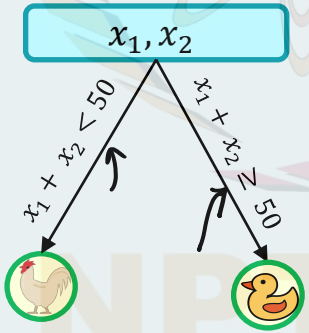
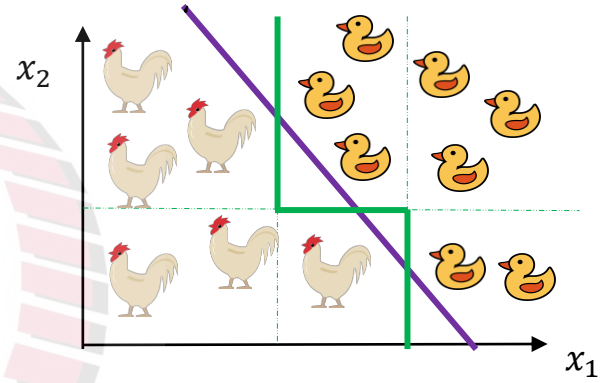


Fig. 2



Task: Classification of birds into duck or hen

Types

- Based on the type of task performed, the decision trees are of two types:

Regression decision trees	Classification decision trees
Used for predicting continuous numerical values or quantities	Used for predicting categorical or discrete values
The outputs in the leaf nodes are real numbers or numeric values	The output in the leaf nodes are class labels or categories
Splitting criteria: Based on minimizing the variance or mean squared error.	Splitting criteria: Based on metrics such as the Gini index or information gain, which measure the impurity or entropy of the data

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Types

- Based on the type of task performed, the decision trees are of two types:

Regression decision trees	Classification decision trees
Examples: CART (Classification and Regression Trees)	Examples: CART (Classification and Regression Trees), ID3 (Iterative Dichotomizer 3)*, and C4.5

*The ID3 algorithm can be used to construct a decision tree for regression by replacing Information Gain with *Standard Deviation Reduction* [https://www.saedsayad.com/decision_tree_reg.htm]

Classification and Regression Trees

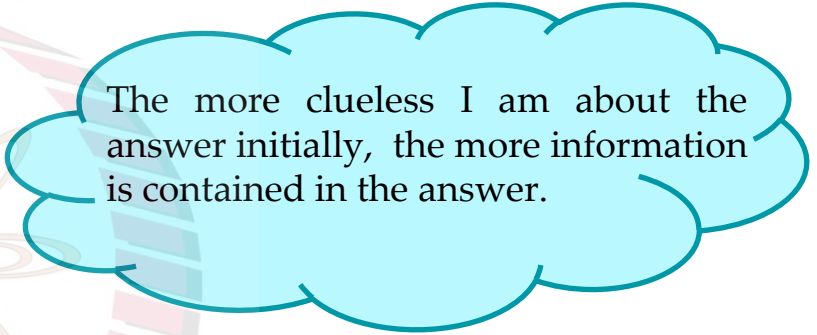
Six types of questions

- Should the attributes be binary or multivalued?
- Which feature or feature combination should be tested at a node?
- If the tree becomes "too large", can it be pruned to make it smaller and simpler?
- If a leaf node is impure, how should a category label be assigned to it?
- How should missing data be handled?

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Splitting Criterion for Classification Trees

- Information answers questions
- The splitting criteria revolve around gaining the maximum possible information from each split.
- For classification trees, the goodness of a split is evaluated using an **impurity measure**.
- A split is considered **pure** when, after the split, all instances that choose a specific branch belong to the same class.



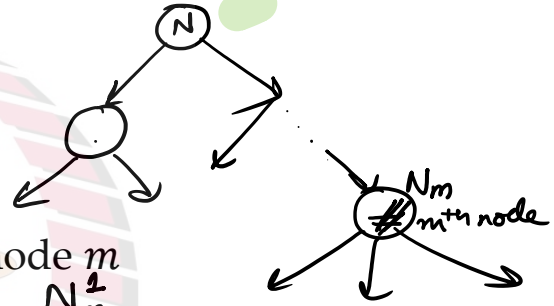
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Splitting Criterion for Classification Trees

Consider,

N: total number of training samples

N_m: the number of training samples reaching node m



N_m^i: the number of training samples of class i reaching node m

$$\sum_{i=0}^{C-1} N_m^i = N_m$$

class 1 $\longrightarrow$ N_m^1

class 0 $\longrightarrow$ N_m^0 $+ = N_m$

- Given, a samples reaches node m, the probability for it to belong to class i is

$$P(i|\boldsymbol{x}, m) = p_m^i = \frac{N_m^i}{N_m}$$

Class-1 $\frac{N_m^1}{N_m} = 0$

Class-0 $\frac{N_m^0}{N_m} = 1$

- Note that node m is pure iff

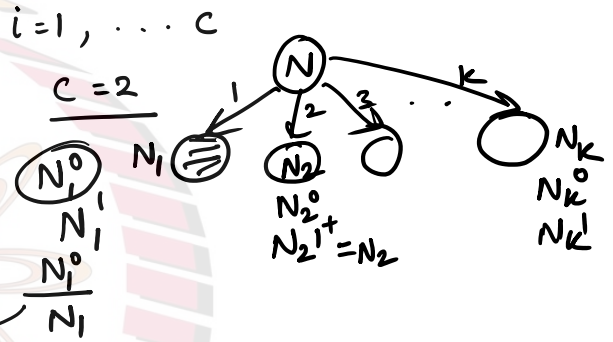
$p_m^i = 1 \; for \; one \; i \; and \; is \; zero \; for \; remaining \; i.$

Entropy- Splitting Criterion for Classification Trees

1. **Entropy**:

$$S_m = -\sum_{i=1}^{C} p_m^i \log_2 p_m^i$$

$$0 \cdot \log_2 0 \equiv 0$$



where,

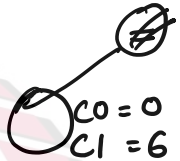
- Let us say that we are at the root node. If the training set N at the root node is divided into k subsets, $N^1, N^2, \ldots, N^k$, the entropy may be reduced.
- The amount of reduction in entropy is the **information gain**, given by

$$Gain(N; N^1, N^2 \ldots N^k) = \mathrm{S(N)} - \sum_{i=1}^{k} \frac{N^i}{N} S(N^i)$$

Entropy - Splitting Criterion for Classification Trees

Entropy:

$$S_m = -\sum_{i=0}^{C-1} p_m^i \log_2 p_m^i$$



Examples:

pure node

C0	0
C1	6

① $p^{C0} = 0/6 = 0$ $p^{C1} = 6/6 = 1$

Entropy = – 0 log 0 – 1 log 1 = – 0 – 0 = 0

C0	1
C1	5

② $p^{C0} = 1/6$ $p^{C1} = 5/6$

Entropy = – (1/6) $\log_2$ (1/6) – (5/6) $\log_2$ (1/6) = 0.65

C1	2
C2	4

③ $p^{C0} = 2/6$ $p^{C1} = 4/6$

Entropy = – (2/6) $\log_2$ (2/6) – (4/6) $\log_2$ (4/6) = 0.92

Gini Index- Splitting Criterion for Classification Trees

2. Gini Index:

$$Gini(m) = 1 - \sum_{i=0}^{C-1} (p_m^i)^2$$

Examples:

(1)

C0	0
C1	6

$p^{C0} = 0/6 = 0 \quad p^{C1} = 6/6 = 1$

Gini = $1 - (p^{C0})^2 - (p^{C1})^2 = 1 - 0 - 1 = 0$

(2)

C0	1
C1	5

$p^{C0} = 1/6 \quad p^{C1} = 5/6$

Gini = $1 - (1/6)^2 - (5/6)^2 = 0.278$

$1 - \left(\frac{1}{6}\right)^2 - \left(\frac{5}{6}\right)^2$

$= 1 - \frac{1}{4} - \frac{1}{4}$

$= 0.5$

(3)

C1	2
C2	4

$p^{C0} = 2/6 \quad p^{C1} = 4/6$

Gini = $1 - (2/6)^2 - (4/6)^2 = 0.444$

Misclassification Error: Splitting Criterion for Classification Trees

3. Misclassification Error:

$$\phi(m) = 1 - \max(p_m^i \; \forall \; i \in \{0, 1, \ldots, C-1\})$$

Examples:

C0	0
C1	6

$p^{C0} = 0/6 = 0 \quad p^{C1} = 6/6 = 1$

Error = 1 – max (0, 1) = 1 – 1 = 0

C0	1
C1	5

$p^{C0} = 1/6 \quad p^{C1} = 5/6$

Error = 1 – max (1/6, 5/6) = 1 – 5/6 = 1/6

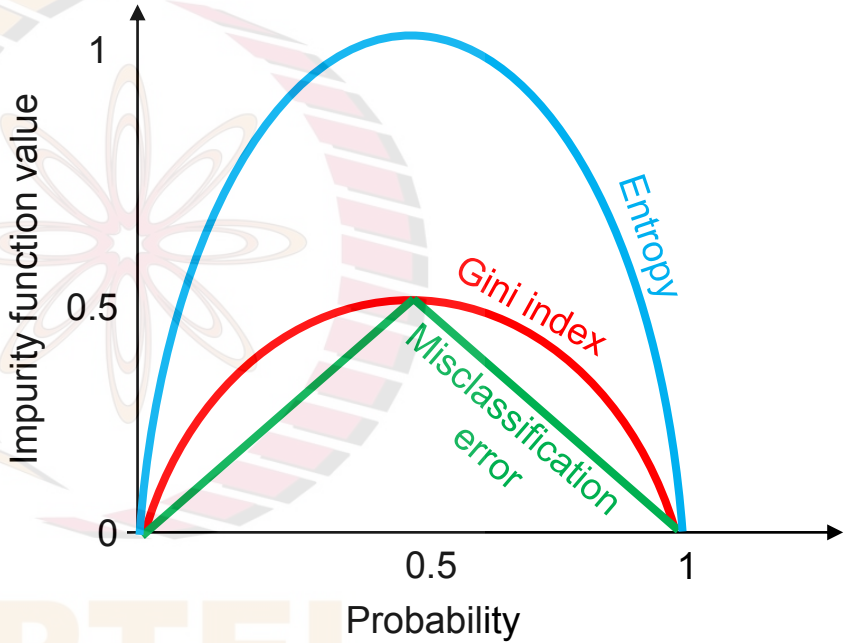
C0	2
C1	4

$p^{C0} = 2/6 \quad p^{C1} = 4/6$

Error = 1 – max (2/6, 4/6) = 1 – 4/6 = 1/3

Two Category Case

- For a two-category case, the impurity functions (Entropy/ Gini/ misclassifications) peak at equal class frequencies.



Pg. 299, Duda, Richard O., and Peter E. Hart. *Pattern classification*. John Wiley & Sons, 2006.

Gain Ratio: Splitting Criterion for Classification Trees

4. Gain Ratio:

S(N) – Split-info
= gain

Consider root node and entropy as the split criterion:

$$GainRatio(N; N^1, N^2 \dots N^k) = \frac{Gain(N; N^1, N^2 \dots N^k)}{SplitINFO(N; N^1, N^2 \dots N^k)}$$

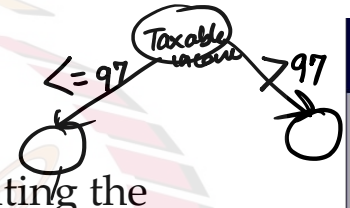
where,

$$SplitINFO(N; N^1, N^2 \dots N^k) = -\sum_{i=1}^{k} \frac{N^i}{N} \log \frac{N^i}{N}$$

$$Gain(N; N^1, N^2 \dots N^k) = \mathrm{S(N)} - \sum_{i=1}^{k} \frac{N^i}{N} S(N^i)$$

Splitting Continuous Attributes

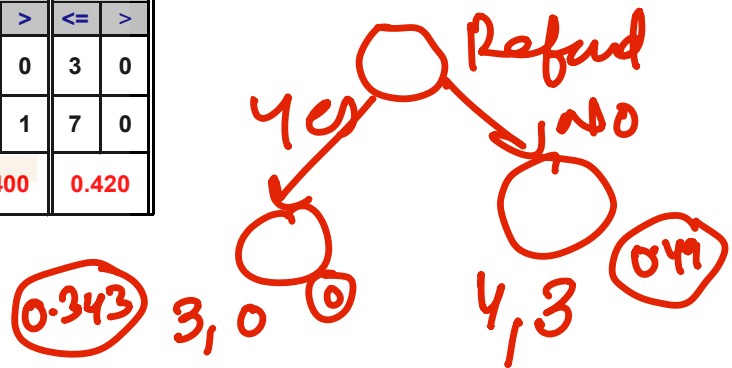
- **Splitting Criteria**: (here) Gini index
- For efficient computation: For each attribute,
  1. Sort the attribute on values
  2. Linearly scan these values, each time updating the count matrix and computing the Gini index
  3. Choose the split position that has the least Gini index



Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Cheat	No	No	No	Yes	Yes	Yes	No	No	No	No
Taxable Income										
Sorted Values	60	70	75	85	90	95	100	120	125	220
Split Positions	55	65	72	80	87	92	97	110	122	172	230
	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >	<= >
Yes	0 3	0 3	0 3	0 3	1 2	2 1	3 0	3 0	3 0	3 0	3 0	3 0
No	0 7	1 6	2 5	3 4	3 4	3 4	3 4	4 3	5 2	6 1	7 0	
Gini	0.420	0.400	0.375	0.343	0.417	0.400	0.300	0.343	0.375	0.400	0.420	

3/10 Cheat
7/10



Reference of Example: Introduction to Data Mining, 2nd ed., Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Bipin Kumar, published by Pearson, 2019, ISBN 978-0-13-312890-1

Splitting Categorical Attributes

- **Splitting Criteria**: (here) Gini index
- For efficient computation: For each attribute,
  1. For each distinct value, gather counts for each class
  2. Use the count matrix to make decisions





Multi-way split

Two-way split
(find best partition of values)

	CarType		
	Family	Sports	Luxury
C1	1	2	1
C2	4	1	1
Gini	0.393		

	CarType	
	{Sports, Luxury}	{Family}
C1	3	1
C2	2	4
Gini	0.400	

	CarType	
	{Sports}	{Family, Luxury}
C1	2	2
C2	1	5
Gini	0.419	

[Assume a dataset with the above count has been provided, and you wish to split the 'CarType' attribute.]

To Summarize

What we learnt in this module is as follows:

- Introduction
- Examples
- Univariate and Multivariate Trees
- Classification Trees

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Reference: **Introduction to machine learning / Ethem Alpaydin**